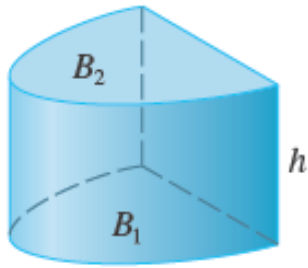
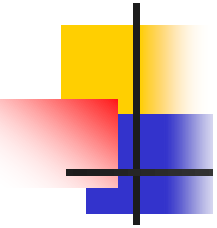
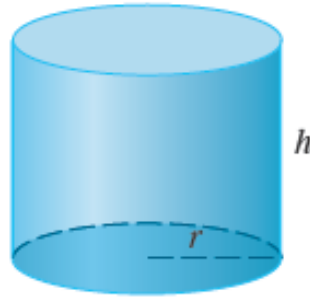


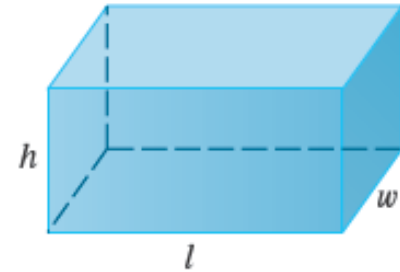
VOLUME (OF SOLIDS OF REVOLUTIONS)



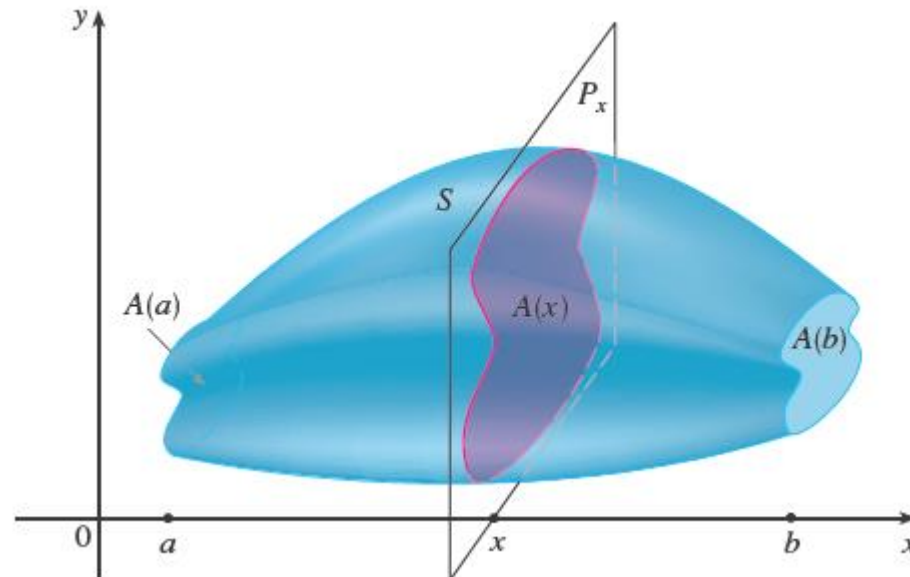
(a) Cylinder $V = Ah$



(b) Circular cylinder $V = \pi r^2 h$



(c) Rectangular box $V = lwh$



Let's divide S into n "slabs" of equal width Δx by using the planes $P_{x_1}, P_{x_2}, \dots$ to slice the solid. (Think of slicing a loaf of bread.) If we choose sample points x_i^* in $[x_{i-1}, x_i]$, we can approximate the i th slab S_i (the part of S that lies between the planes $P_{x_{i-1}}$ and P_{x_i}) by a cylinder with base area $A(x_i^*)$ and "height" Δx . (See Figure 3.)

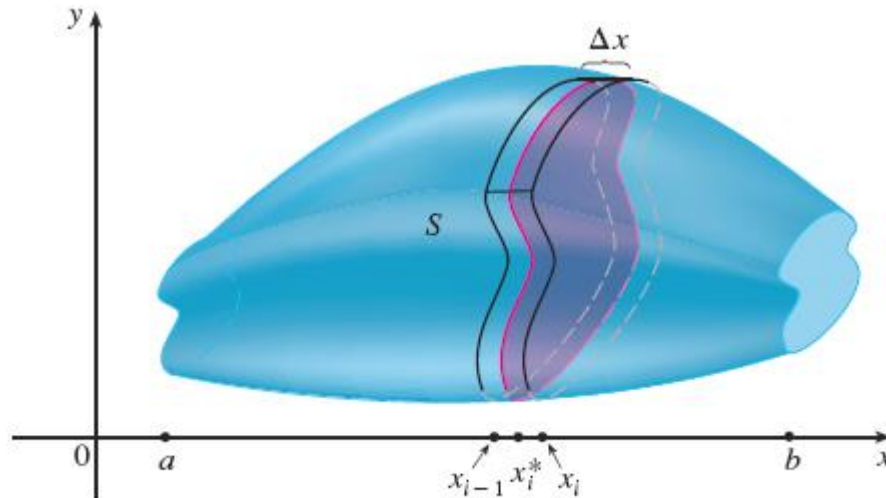
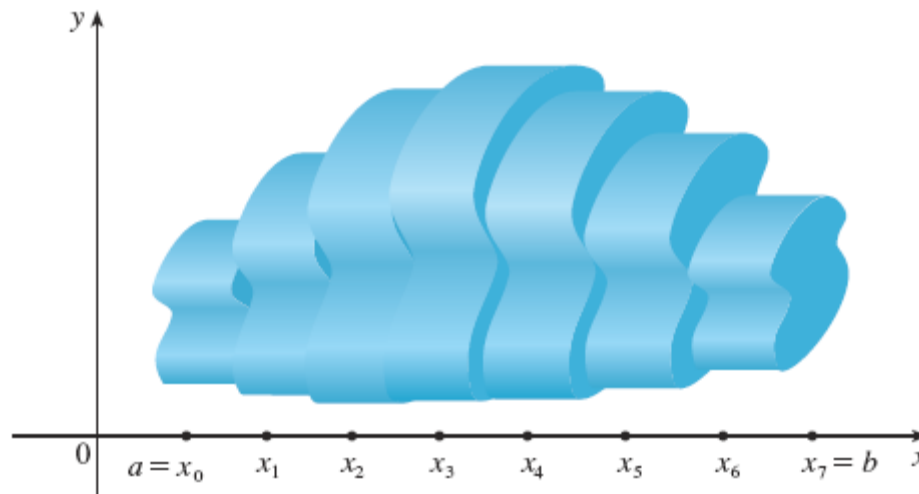


FIGURE 3



The volume of this cylinder is $A(x_i^*) \Delta x$, so an approximation to our intuitive conception of the volume of the i th slab S_i is

$$V(S_i) \approx A(x_i^*) \Delta x$$

Adding the volumes of these slabs, we get an approximation to the total volume (that is, what we think of intuitively as the volume):

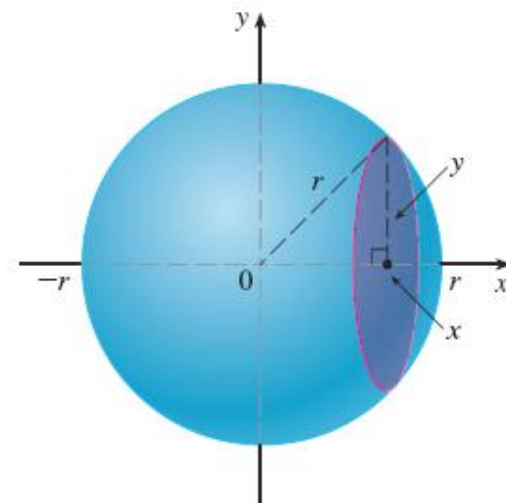
$$V \approx \sum_{i=1}^n A(x_i^*) \Delta x$$

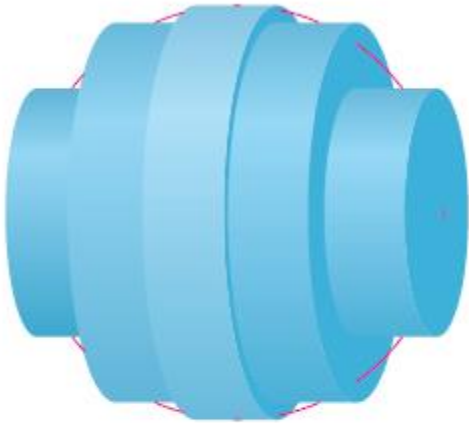
This approximation appears to become better and better as $n \rightarrow \infty$. (Think of the slices as becoming thinner and thinner.) Therefore we *define* the volume as the limit of these sums as $n \rightarrow \infty$. But we recognize the limit of Riemann sums as a definite integral and so we have the following definition.

Definition of Volume Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x , through x and perpendicular to the x -axis, is $A(x)$, where A is a continuous function, then the **volume** of S is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$$

EXAMPLE 1 Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.





(a) Using 5 disks, $V \approx 4.2726$



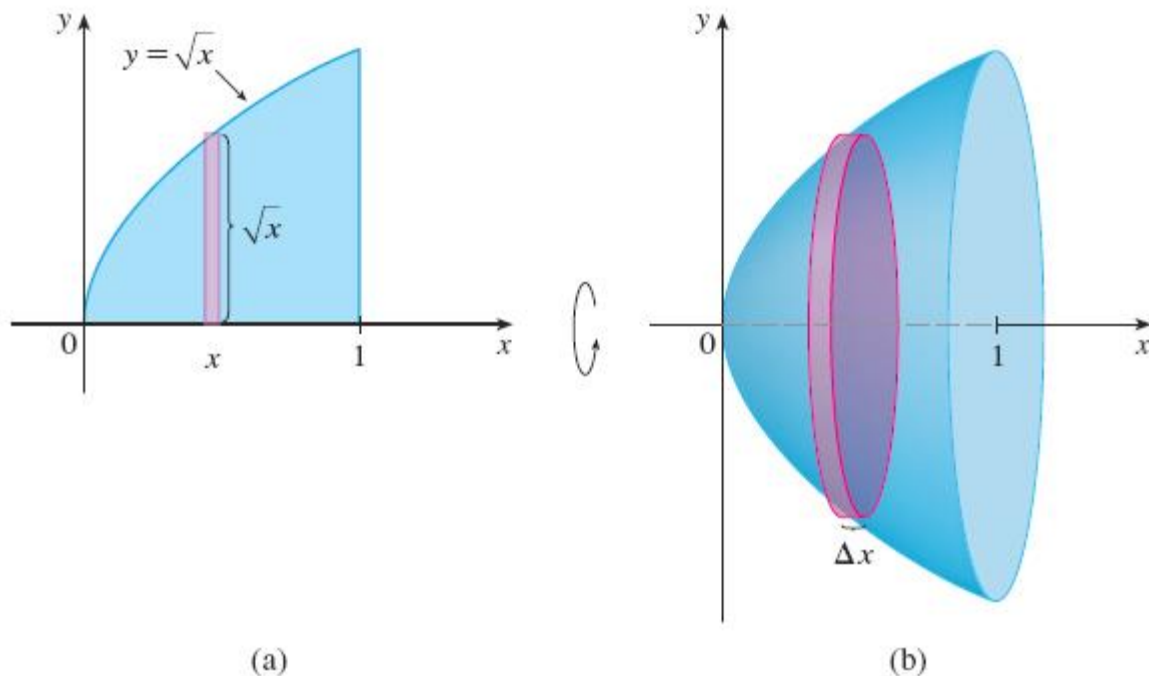
(b) Using 10 disks, $V \approx 4.2097$



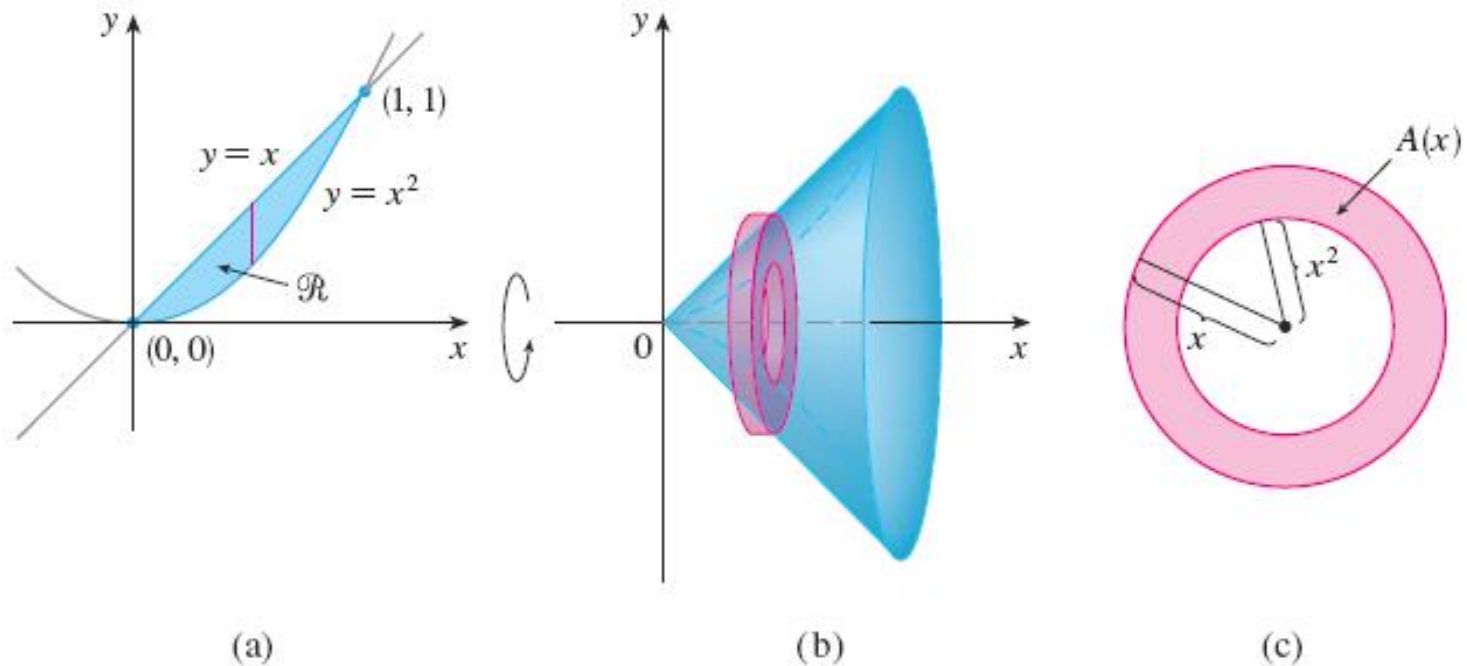
(c) Using 20 disks, $V \approx 4.1940$

Actual value=4.18879

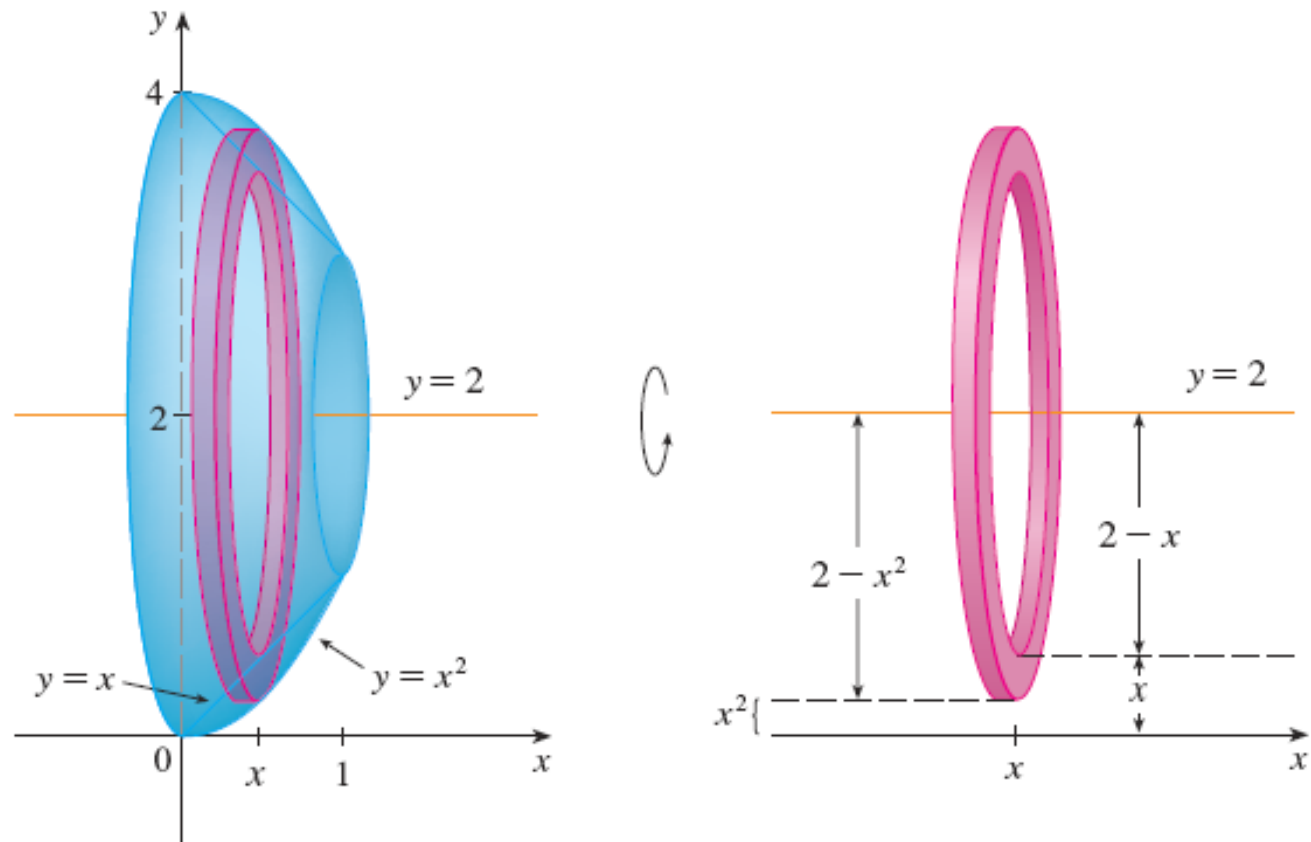
EXAMPLE 2 Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \sqrt{x}$ from 0 to 1. Illustrate the definition of volume by sketching a typical approximating cylinder.



EXAMPLE 4 The region $\mathcal{R}$ enclosed by the curves $y = x$ and $y = x^2$ is rotated about the x -axis. Find the volume of the resulting solid.



EXAMPLE 5 Find the volume of the solid obtained by rotating the region in Example 4 about the line $y = 2$.



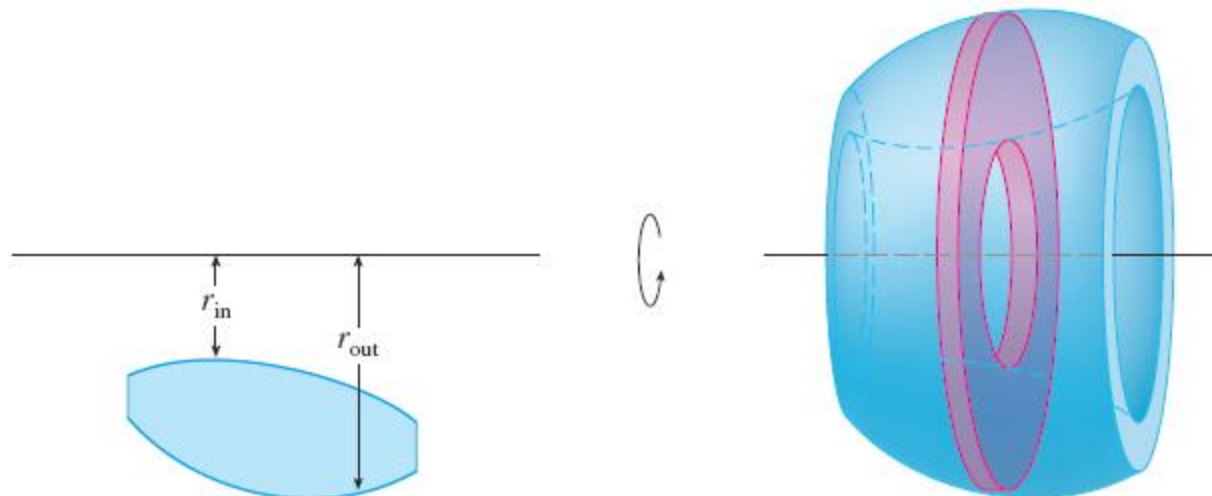
The solids in the Examples are all called **solids of revolution** because they are obtained by revolving a region about a line. In general, we calculate the volume of a solid of revolution by using the basic defining formula

$$V = \int_a^b A(x) dx \quad \text{or} \quad V = \int_c^d A(y) dy$$

and we find the cross-sectional area $A(x)$ or $A(y)$ in one of the following ways:

- If the cross-section is a disk, we find the radius of the disk (in terms of x or y) and use
$$A = \pi(\text{radius})^2$$
- If the cross-section is a washer, we find the inner radius r_{in} and outer radius r_{out} from a sketch and compute the area of the washer by subtracting the area of the inner disk from the area of the outer disk:

$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$$



Multiple integrals

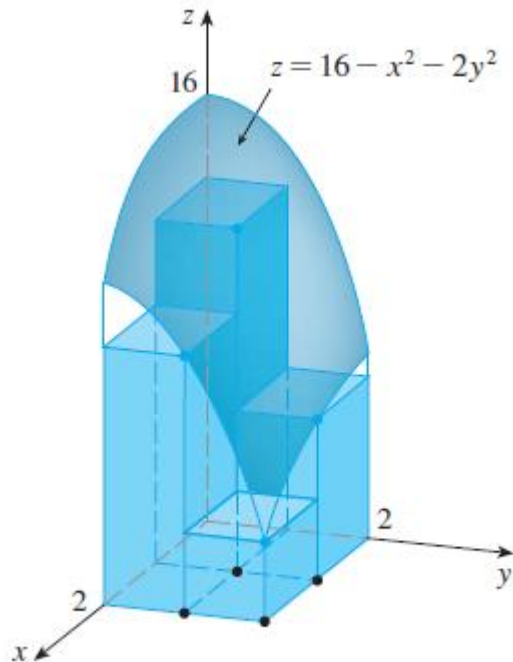


FIGURE 7

Figure 11 shows the solid whose volume is calculated in Example 2. It lies above the xy -plane, below the paraboloid $z = x^2 + y^2$, and between the plane $y = 2x$ and the parabolic cylinder $y = x^2$.

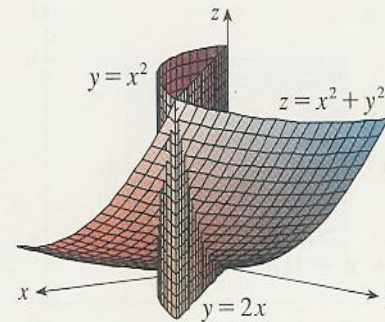


FIGURE 11

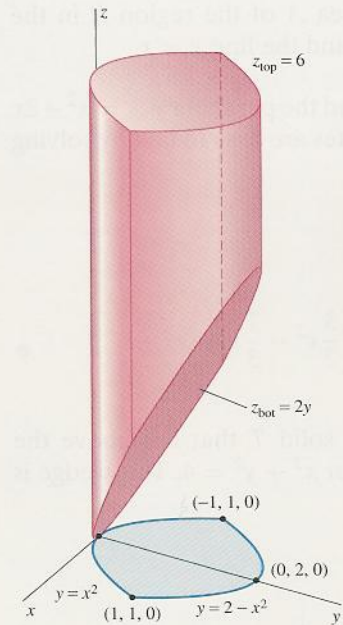
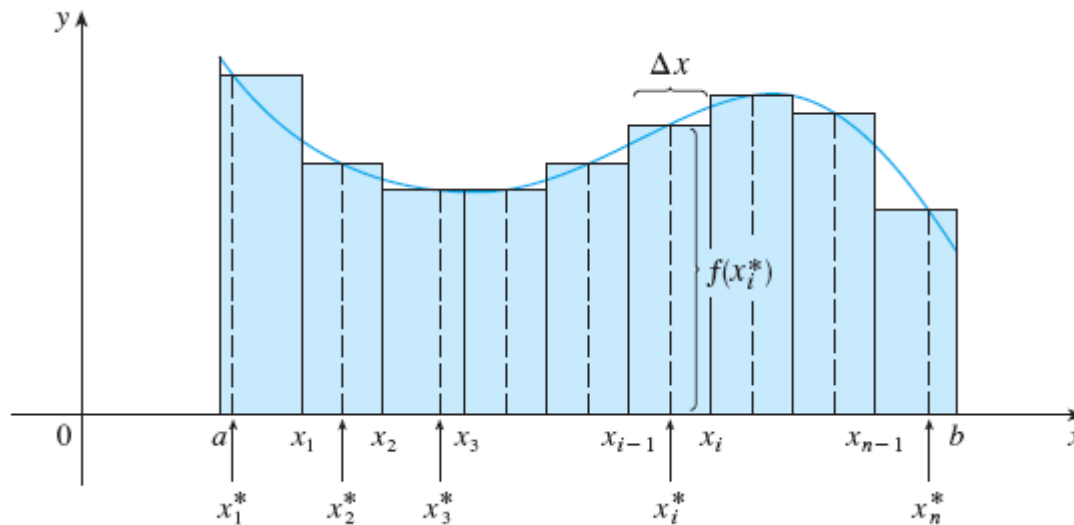


FIGURE 14.3.12 The solid T of Example 4.

Review of Definite Integral



2

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

Double integrals over Rectangles

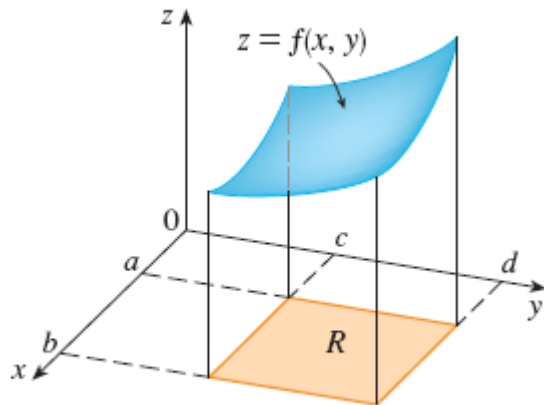
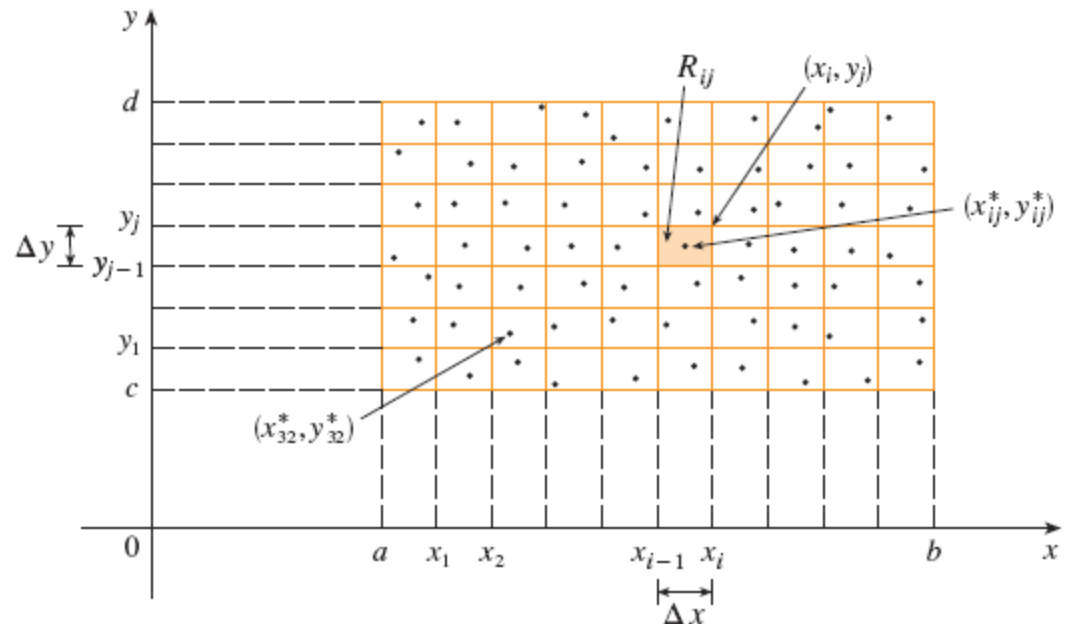


FIGURE 2



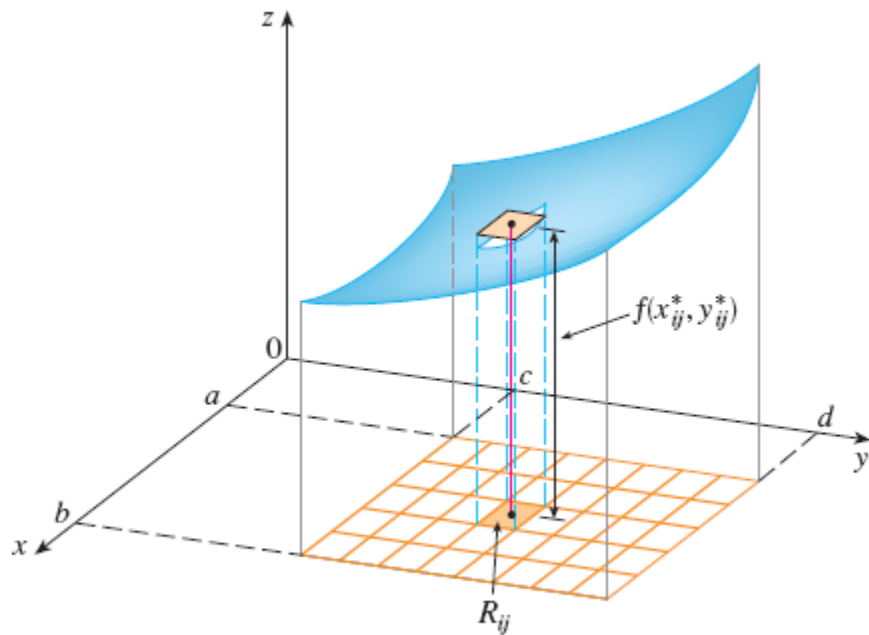


FIGURE 4

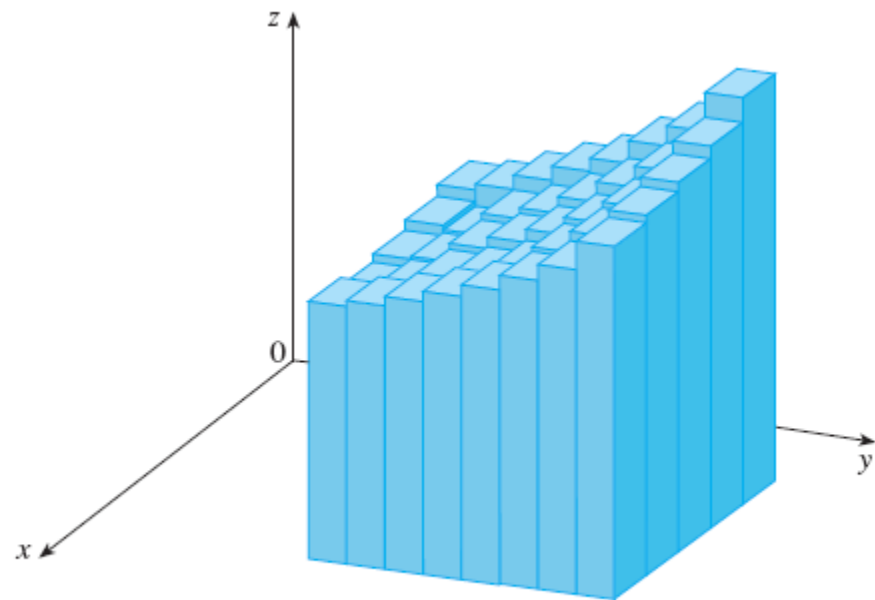


FIGURE 5

Double integrals as Volumes

4

$$V = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

5 **Definition** The **double integral** of f over the rectangle R is

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

if this limit exists.

The precise meaning of the limit in Definition 5 is that for every number $\varepsilon > 0$ there is an integer N such that

$$\left| \iint_R f(x, y) dA - \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A \right| < \varepsilon$$

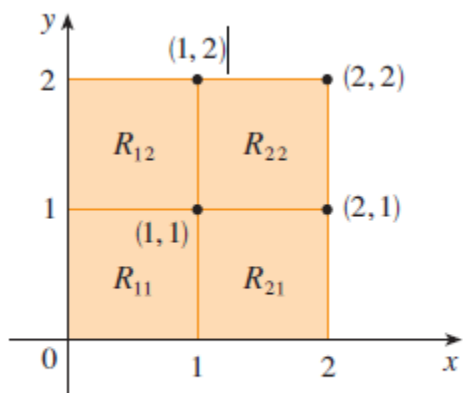
for all integers m and n greater than N and for any choice of sample points (x_{ij}^*, y_{ij}^*) in R_{ij} .

6

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) \Delta A$$

If $f(x, y) \geq 0$, then the volume V of the solid that lies above the rectangle R and below the surface $z = f(x, y)$ is

$$V = \iint_R f(x, y) dA$$



EXAMPLE 1 Estimate the volume of the solid that lies above the square $R = [0, 2] \times [0, 2]$ and below the elliptic paraboloid $z = 16 - x^2 - 2y^2$.

FIGURE 6

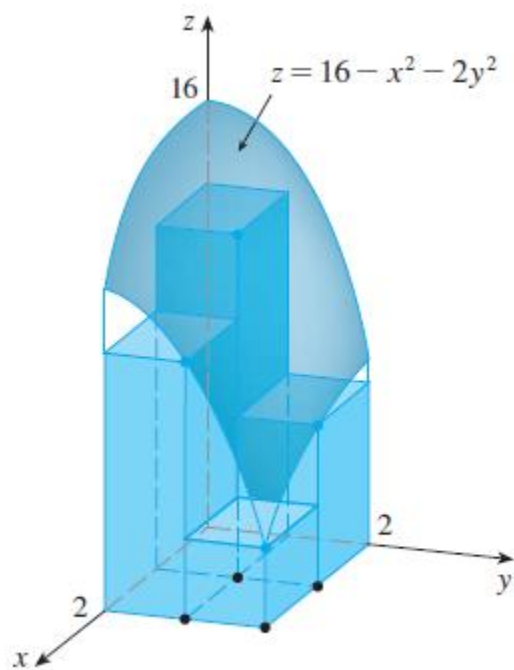
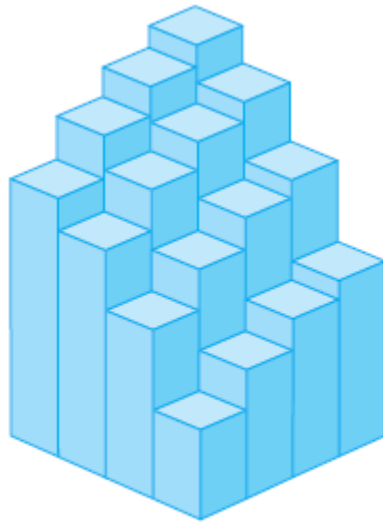
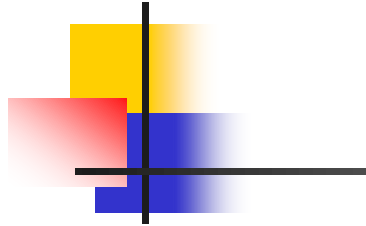
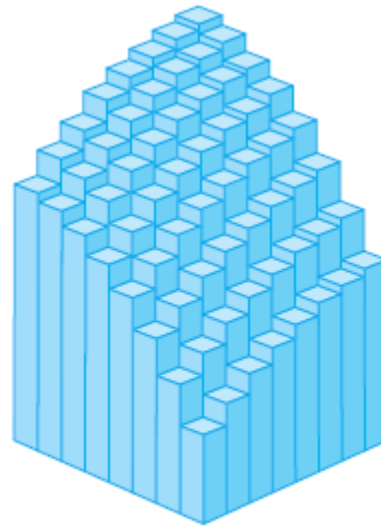


FIGURE 7

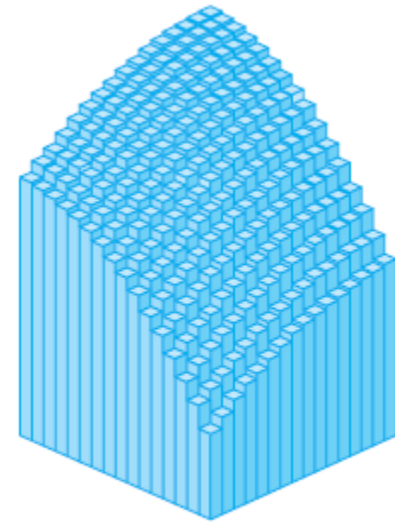
$$\begin{aligned}
 V &\approx \sum_{i=1}^2 \sum_{j=1}^2 f(x_i, y_j) \Delta A \\
 &= f(1, 1) \Delta A + f(1, 2) \Delta A + f(2, 1) \Delta A + f(2, 2) \Delta A \\
 &= 13(1) + 7(1) + 10(1) + 4(1) = 34
 \end{aligned}$$



(a) $m = n = 4$, $V \approx 41.5$



(b) $m = n = 8$, $V \approx 44.875$



(c) $m = n = 16$, $V \approx 46.46875$

exact volume is 48.

EXAMPLE 2 If $R = \{(x, y) \mid -1 \leq x \leq 1, -2 \leq y \leq 2\}$, evaluate the integral

$$\iint_R \sqrt{1 - x^2} \, dA$$

